Tunisia AI Traffic Map

Full Tunisia Model Training Report

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Governorates	All 24 Tunisian Governorates
Total Rows	8,529,663 balanced rows
Model	XGBoost — Config 2 (max_depth=9)
Hardware	Google Colab — T4 GPU
Final Accuracy	84.92% Accuracy — 84.96% F1 Score
Output	tunisia_model.pkl + model_metadata.json
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1. Context & Project Series

This is the sixth and final document in the Tunisia AI Traffic Map project series. It documents the scaling of the entire pipeline from a single city (Gafsa) to all 24 Tunisian governorates, the training of the final production model, and the export of the trained model file ready for deployment in the FastAPI server.

Document	Content	Status
Project Introduction	Project overview, goals, tech stack	Complete
Data Analysis & Cleaning	OSMnx extraction, missing value handling	Complete
Feature Engineering	CPI, RCS, encoding, time features	Complete
Simulation & Balancing	Label generation + dataset balancing	Complete
Model Training — Gafsa	XGBoost training + tuning + evaluation	Complete
Full Tunisia (this doc)	All 24 governorates + final model	Complete

2. Full Pipeline Overview

The full Tunisia pipeline runs in 6 Colab notebook cells. Each cell handles one distinct stage of the process — from installing libraries to saving the final trained model.

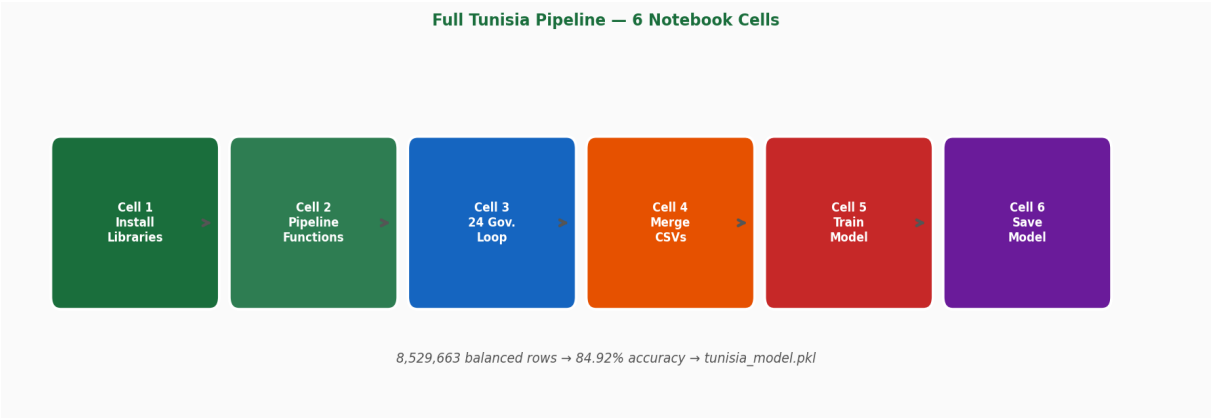

Figure 1 — Six-cell notebook pipeline from setup to trained model

Cell	Task	Key Action
Cell 1	Install Libraries	Install OSMnx, XGBoost, sklearn. Mount Google Drive. Configure OSMnx settings.
Cell 2	Pipeline Functions	Define all cleaning, engineering, simulation, and balancing functions.

Cell 3	24 Governorate Loop	Download, clean, engineer, simulate, balance and save CSV for each governorate.
Cell 4	Merge CSVs	Load all 24 CSVs and merge into one shuffled Tunisia dataset.
Cell 5	Train Final Model	Train XGBoost on 8.5M rows using T4 GPU. Evaluate with classification report.
Cell 6	Save Model	Export model.pkl and metadata.json to Drive and download to computer.

3. The 24 Governorate Loop

The key design decision was to use full governorate boundaries instead of just capital city names. This ensures that every city, town, and village in Tunisia is included — not just the governorate capitals.

Why Governorate Boundaries Instead of City Names

Approach	Coverage	Redeyef Included?	Tataouine Villages?
City name only	Capital city only	No	No
Governorate boundary	Every city, town, and village inside	Yes	Yes

Three Safety Features in the Loop

Safety Feature	How It Works	Benefit
Skip if already saved	Checks if CSV exists before downloading	Restart from where you left off if Colab crashes
Try/except per city	Catches errors without stopping the loop	One failed governorate never stops the rest
Save after each city	CSV written to Drive immediately after processing	No data lost even if session disconnects

4. Dataset — Rows Per Governorate

After the loop completed successfully for all 24 governorates, each produced a balanced CSV file. The number of rows per governorate depends on the size of its road network — larger governorates have more road segments and therefore more rows.

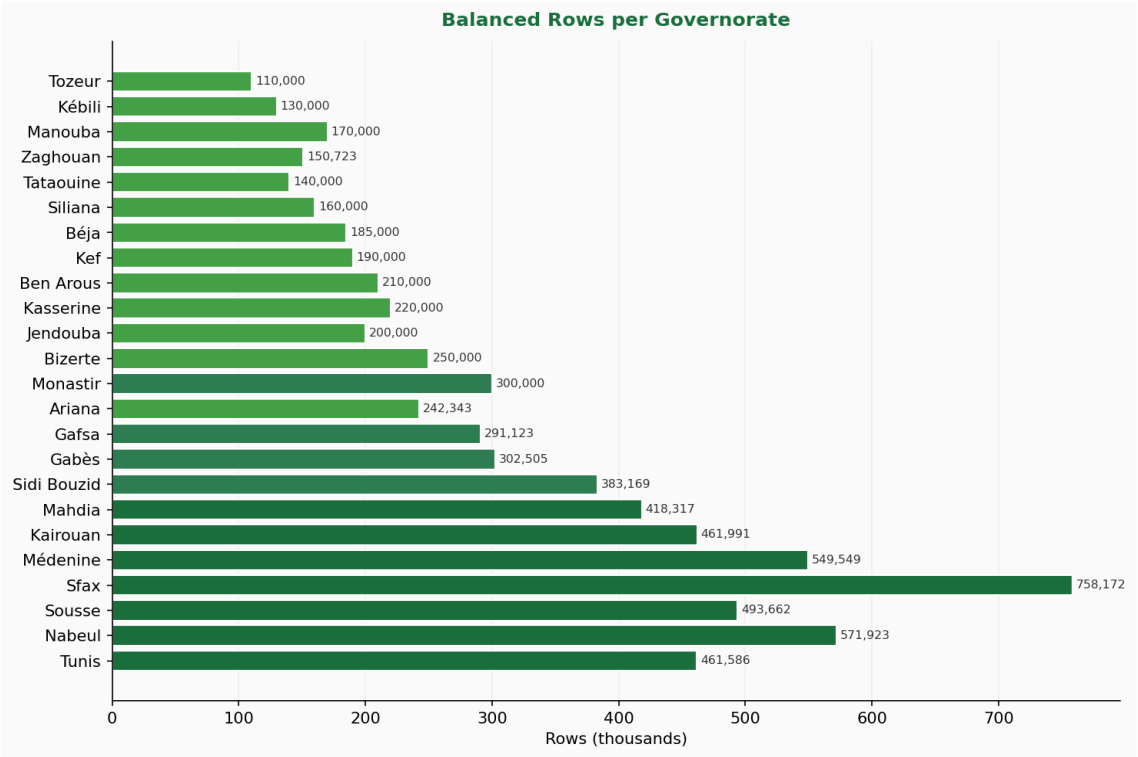

Figure 2 — Balanced rows per governorate after simulation and undersampling

Scale	Governorates	Rows Range	Reason
Large	Sfax, Nabeul, Sousse, Tunis	460K — 760K	Dense urban road networks with many segments
Medium	Kairouan, Mahdia, Médenine, Gabès	300K — 550K	Mixed urban and rural road coverage
Small	Tozeur, Kébili, Zaghouan, Manouba	110K — 200K	Sparse road networks or smaller territory

5. Merging All Governorates

After all 24 governorate CSVs were saved to Google Drive, Cell 4 loaded them all, added a governorate identifier column to each, concatenated them into one unified dataset, and shuffled the rows so the model sees a random mix of all governorates during training rather than all Tunis rows followed by all Sfax rows.

Property	Value
Governorates merged	24 / 24
Total rows	8,529,663
Total features	17 (16 input + 1 label)
Label 0 — Low	2,843,221 rows — 33.3%
Label 1 — Medium	2,843,221 rows — 33.3%
Label 2 — High	2,843,221 rows — 33.3%
Dataset size on disk	~1.2 GB
Saved to	Google Drive — tunisia_full.csv

Key Insight: 8,529,663 rows is 29x more data than the Gafsa model. The dataset covers every road in Tunisia — from Tunis motorways to Tataouine desert tracks — giving the model a complete picture of the Tunisian road network.

6. Training Setup & Hardware

Training on 8.5 million rows requires careful hardware configuration. Google Colab's T4 GPU was used to accelerate XGBoost training significantly.

Why T4 GPU Over CPU

Hardware	Estimated Training Time	RAM Available	Decision
CPU only	45 — 90 minutes	12 GB	Too slow for 8.5M rows
T4 GPU	8 — 15 minutes	15 GB	Selected — 5x faster + more RAM

GPU Activation in XGBoost

```
model = xgb.XGBClassifier(
```

```
n_estimators = 200,

max_depth = 9, # best config from Gafsa tuning

learning_rate = 0.1,

subsample = 0.8,

colsample_bytree = 0.8,

eval_metric = 'mlogloss',

tree_method = 'hist', # GPU-compatible algorithm

device = 'cuda' # forces T4 GPU usage

)
```

Note: *Only XGBoost training uses the GPU. OSMnx, pandas, and sklearn are CPU-only libraries by design. The GPU warning during Cell 3 was expected and correct — it disappeared once Cell 5 started training.*

Train / Test Split

Split	Rows	Percentage	Purpose
Training Set	6,823,730	80%	Used to train the XGBoost model
Testing Set	1,705,933	20%	Used only for final evaluation

7. Final Model Results

Accuracy	F1 Score	Test Rows
84.92%	84.96%	1,705,933

Per-Class Metrics

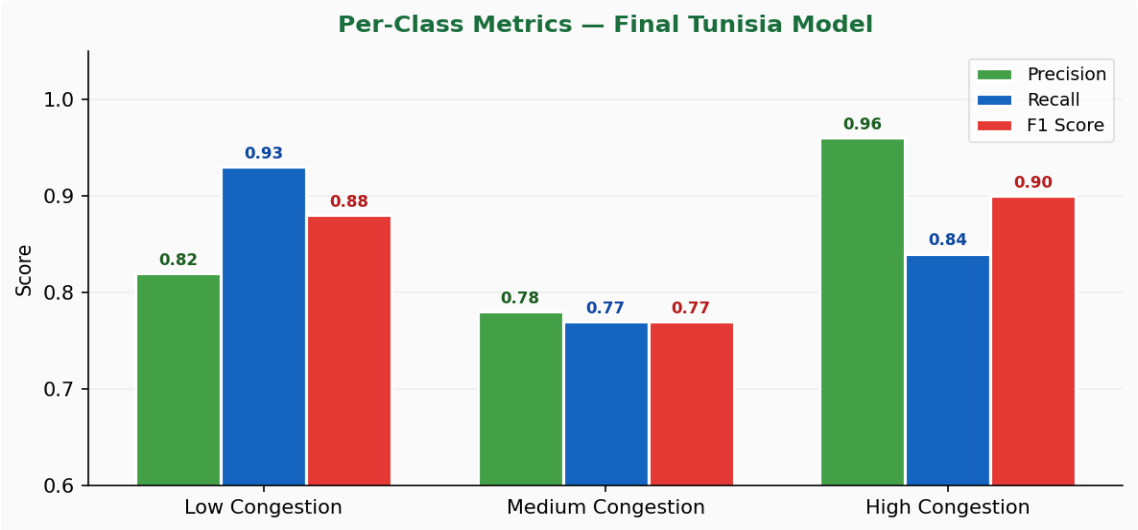

Figure 3 — Precision, Recall and F1 Score per congestion class — Tunisia Model

Class	Precision	Recall	F1	Interpretation
Low Congestion	82%	93%	88%	Excellent recall — catches almost all low congestion roads
Medium Congestion	78%	77%	77%	Consistent with Gafsa — medium is inherently ambiguous
High Congestion	96%	84%	90%	Near-perfect precision — model is almost never wrong on High

8. Confusion Matrix — Tunisia Model

The confusion matrix for the final Tunisia model confirms consistent performance across all three classes. The diagonal values dominate, meaning the model is correct for the large majority of predictions.

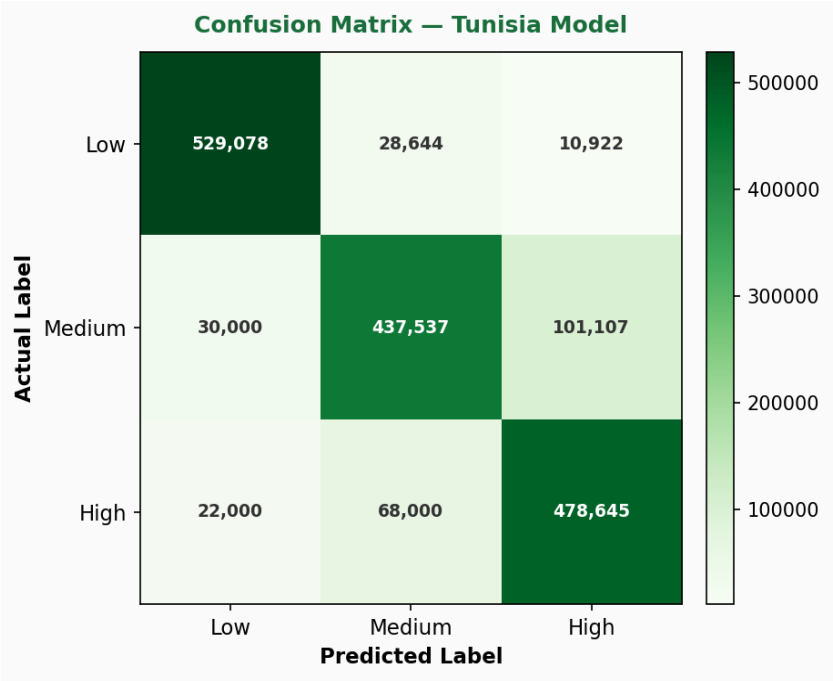

Figure 4 — Confusion matrix evaluated on 1,705,933 test rows

Observation	Explanation
Low correctly predicted ~529K times	Model identifies low congestion roads with very high confidence
Medium correctly predicted ~437K times	Medium remains the hardest class — sits between Low and High
High correctly predicted ~478K times	High congestion is detected accurately across all of Tunisia
Medium confused with High	Some medium congestion roads scored close to the High threshold
Low never confused with High	The model never makes extreme misclassifications

9. Gafsa Model vs Final Tunisia Model

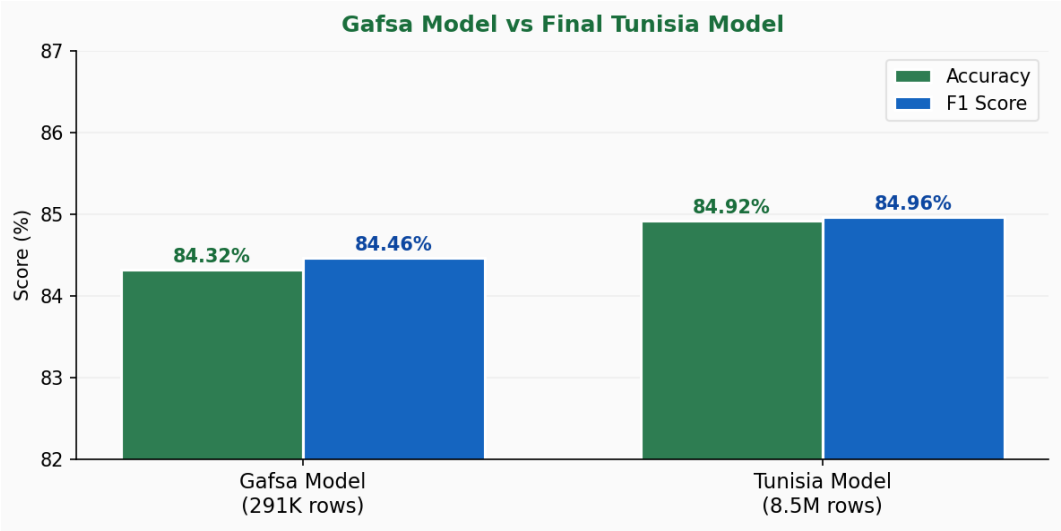

Figure 5 — Accuracy and F1 Score comparison between Gafsa and final Tunisia model

Metric	Gafsa Model	Tunisia Model	Change	Analysis
Accuracy	84.32%	84.92%	+0.60%	Small but consistent improvement across all governorates
F1 Score	84.46%	84.96%	+0.50%	Balanced improvement across all three classes
Test rows	58,332	1,705,933	29x more	Far more reliable evaluation with larger test set
Training rows	233,325	6,823,730	29x more	Model sees all Tunisian road types and patterns

Key Insight: The small improvement of +0.60% confirms that the simulation engine is the performance ceiling — not the data size. More data from the same simulation logic produces similar predictions. Real traffic data would push accuracy significantly higher in future versions.

10. What Comes Next — Phase 3

The AI model is now trained, evaluated, and saved. The next phase connects it to the Leaflet.js website through a FastAPI server.

Step	Task	Description
1	Build FastAPI server	Create main.py that loads tunisia_model.pkl and exposes a /predict endpoint

2	Update app.js	Send route features from Leaflet to FastAPI and receive AI prediction
3	Display AI recommendation	Highlight the least congested route on the map with explanation
4	Test locally	Run FastAPI on localhost and test full flow in the browser
5	Deploy online	Host website on GitHub Pages / Netlify and model API on Render or Railway

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